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A quadruplex real-time qPCR assay for the simultaneous assessment of total human DNA, human male DNA, DNA degradation and the presence of PCR inhibitors in forensic samples: A diagnostic tool for STR typing

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Abstract

A quadruplex real-time qPCR assay was developed to simultaneously assess total human DNA, human male DNA, DNA degradation and PCR inhibitors in forensic samples. Specifically, the assay utilizes a ~170–190 bp target sequence that spans the TH01 STR locus to quantify total human DNA (nuTH01), a 137 bp target sequence directly adjacent to the SRY gene to quantify human male DNA (nuSRY), a 67 bp target sequence flanking the CSF1PO STR locus (nuCSF) to assess degradation (nuCSF:nuTH01 ratio) and a 77 bp synthetic DNA template used as an internal PCR control target sequence (IPC) for the assessment of PCR inhibition. Validation studies, performed on an ABI 7500 SDS instrument using TaqMan® and TaqManMGB® detection, indicate each of the targets in the quadruplex assay performs effectively and is informative even when challenged with DNase-degraded and hematin-inhibited samples. The nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay is envisioned to assist in the choice of the most informative DNA typing system available, which may include standard autosomal STR typing when the results indicate the presence of non-degraded, single gender DNA or non-degraded, male:female mixtures at ratios expected to yield probative alleles; Y STR typing in samples containing a male component that is overwhelmed by the presence of an excess of female DNA; reduced amplicon size STR typing (''MiniSTRs'') where the nuCSF:nuTH01 ratio indicates the sample is highly degraded; enhanced STR amplification with additional AmpliTaq Gold®/BSA and/or sample clean-up when the presence of PCR inhibitors is suggested by a delayed IPC C_T value or mitochondrial DNA typing in samples where little to no nuclear DNA is detected. The present study includes evaluations of species specificity, sensitivity, precision, reproducibility, male–female mixtures, population samples and applications to various casework-type samples as indicated by the Scientific Working Group on DNA Analysis Methods (SWGDAM) guidelines.
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1. Introduction

The increasing number of DNA typing systems and decreasing amount of DNA per evidence item submitted to forensic laboratories has driven the development of several real-time quantitative PCR (qPCR) assays capable of simultaneously estimating total human DNA and male DNA or of estimating total human DNA and extent of DNA degradation [1–5]. While standard autosomal short tandem repeat (STR) typing with the 13 core CODIS loci is generally the most informative typing system for the vast majority of samples encountered in forensic DNA casework, a significant number of samples submitted to forensic DNA laboratories are better suited for typing with reduced size amplicons (''MiniSTRs'') [6], Y-chromosome-specific STRs [7–9] or mitochondrial DNA typing [10–13]. A method that would allow for the simultaneous determination of total human DNA present, the total male DNA present, the degree of degradation in the sample and the presence of inhibitors would allow for further streamlining of the DNA analysis process and more prudent use of a limited sample. While the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay was developed for use with STR typing systems, alternate typing methodologies such as single nuclear polymorphisms (SNPs) [14], Alu repeats [15], whole genome

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amplifications (WGA) [16] and mitochondrial DNA typing may also benefit from the advanced knowledge of the condition and composition of the sample.

We have previously described a triplex real-time qPCR assay that can simultaneously assess the quantity of total human DNA, the level of degradation and the presence of inhibitors in a forensic sample [17,18]. In this article, we report the developmental validation of a quadruplex real-time qPCR assay for the concurrent assessment of the total human DNA, human male DNA, DNA degradation and the presence of PCR inhibitors in forensic samples. The assay utilizes a ~170–190 bp target sequence that spans the TH01 STR locus (nuTH01) to quantify the total human DNA, a 137 bp target sequence directly adjacent to the SRY gene (nuSRY) to quantify the human male DNA, a 67 bp target sequence flanking the CSF1PO STR locus (nuCSF) to assess degradation (nuCSF quantity/nuTH01 quantity) and a 77 bp synthetic DNA template used as an internal PCR control (IPC) for the assessment of PCR inhibition. This assay was tested, in accordance with the current validation guidelines [19], for species specificity, sensitivity, stability, reproducibility, case-type samples, population distribution of genetic markers, mixed source samples, precision and accuracy.

2. Materials and methods

2.1. qPCR assay target sequences

The quadruplex qPCR assay was developed by adding a Y-chromosome-specific assay to our previously described triplex qPCR assay [17,18], which included the nuTH01, nuCSF and IPC assays. In particular, a 137 bp target sequence directly adjacent to the *SRY* gene (nuSRY) is included to quantify human male DNA. The nuSRY target sequence begins 18 bases upstream (5′) of the coding region of the *SRY* gene (GenBank locus L08063 (HUMSRYZ)) [20]. For each portion of the nuTH01–nuSRY–nuCSF–IPC quadruplex assay, the formation of PCR product is detected in real-time using 5′-hydrolysis chemistry with dual-labeled probes (e.g., TaqMan® or TaqManMGB® probes) (Table 1).

2.2. Standards and samples

Pre-quantified, high molecular weight, human genomic DNA obtained from Promega, Madison, WI (Male #G1471) was used as a qPCR quantification standard. Pre-quantified DNA extracts obtained from Applied Biosystems, Foster City, CA (TaqMan® Control DNA, genomic DNA standards from the Quantifiler™ and QuantiBlot™ kits), Promega (Female #G1521), and ATCC, Manassas, VA (HL60) were also used as control samples in several studies.

Unless otherwise indicated, DNA extracts from reference, non-probative and simulated casework samples were obtained using the California Department of Justice casework organic extraction protocol (ProK/SDS digestion, phenol/chloroform extraction, then Centricon100 (Millipore) concentration into Tris–EDTA buffer (TE^{-4})). For population study samples, whole blood samples were extracted with the QIAMP 96 Blood Kit (Qiagen). For simulated sexual assault samples, differential extractions of sperm and epithelial fractions were performed using a DTT-based digestion protocol [21].

The hematin model system was used to study the effects of inhibition on the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay. Eleven tubes, each containing 40 ng of TaqMan® Control Human Genomic DNA (Applied Biosystems), were spiked with increasing volumes of 1 mM porcine hematin (Sigma, #H3281) and brought to a total volume of 40 μL with sterile water such that the final hematin concentrations ranged from 0 to 800 μM.

The ability of the assay to assess DNA degradation was tested with DNase I fragmented samples as previously described [17]. Male and female degradation series were prepared by fragmenting HL60 female DNA (ATCC) and Raji male DNA (Applied Biosystems Quantifiler Y Standard) with varying ratios of DNase I (Invitrogen, Carlsbad, CA) to DNA (4.4×10^{-5} to 7.1×10^{-5} U/ng) for lengths of time that varied from 1 to 92 min. The level of DNA fragmentation in each sample was assessed by gel electrophoresis (2% agarose, ethidium bromide detection) using Ready-Load λ-DNA/Hind III (Invitrogen) as a size standard. A final degradation series was created by selecting six time-points with varying degrees of

Table 1
Oligonucleotide sequences for nuTH01, nuSRY and nuCSF primers and probes as well as IPC primers, probe and single-stranded oligonucleotide

Oligonucleotide	Sequence, 5′–3′	GenBank accession	Position
nuTH01-F	AGG GTA TCT GGG CTC TGG	D00269	1116–1133
nuTH01-R	GGC TGA AAA GCT CCC GAT TAT	D00269	1281–1301
nuTH01-probe	***Cy5***-ATT CCC ATT GGC CTG TTC CTC CCT T-***BHQ 3***	D00269	1138–1162
nuSRY-F	TGG CGA TTA AGT CAA ATT CGC	L08063	2698–2718
nuSRY-R	CCC CCT AGT ACC CTG ACA ATG TAT T	L08063	2810–2834
nuSRY-probe	***VIC***-CCC TGC TGA TCT GCC T-***MGB-NFQ***	L08063	2750–2765
nuCSF-F	GGG CAG TGT TCC AAC CTG AG	X14720	11,836–11,855
nuCSF-R	GAA AAC TGA GAC ACA GGG TGG TTA	X14720	11,879–11,902
nuCSF-probe	***FAM***-CAA CCT GCT AGT CCT T-***MGB-NFQ***	X14720	11,862–11,877
IPC-F	AAG CGT GAT ATT GCT CTT TCG TAT AG	N/A	N/A
IPC-R	ACA TAG CGA CAG ATT ACA ACA TTA GTA TTG	N/A	N/A
IPC-probe	***NED***-TAC CAT GGC AAT GCT-***MGB-NFQ***	N/A	N/A
IPC-oligo	AAG CGT GAT ATT GCT CTT TCG TAT AGT TAC CAT	N/A	N/A
	GGC AAT GCT TAG AAC AAT ACT AAT GTT GTA ATC TGT CGC TAT GT	N/A	N/A

fragmentation. A "0 min" sample of intact DNA that contained all reaction components for digestion except DNase I was treated identically to the samples in the degradation series. A 1:100 dilution was made of each sample in the series to yield a ~1 ng/μL solution and 4 μL of each was assayed via qPCR. Additionally, mixtures of lightly degraded (fragments > 2072 bp), moderately degraded (fragments < 2027 bp) and highly degraded (fragments < 564 bp) male and female DNA were evaluated at 1:1 and 1:10 ratios of male to female DNA.

Non-human DNA samples (sex of higher organisms unknown) were purchased as pre-quantified genomic DNA extracts from several vendors: *Escherichia coli*, *Clostridium perfringens* (Sigma, St. Louis, MO), *Bacillus subtilis*, *Staphylococcus epidermidis*, *Candida albicans* (ATCC, Manassas, VA), mouse (Promega) and cat, chicken, cow, dog, fish, horse, monkey, pig and rat (Zyagen Labs, San Diego, CA). Additionally, whole blood samples from known male specimens of common domesticated mammals were kindly provided by the following sources: cat, dog (Muller Veterinary Hospital, Walnut Creek, CA), cow, horse (Veterinary Genetics Laboratory, University of California, Davis, CA) and pig (Swine Teaching and Research Facility, University of California, Davis, CA). These five samples were extracted using the California Department of Justice casework organic extraction protocol and the quantity of DNA in each sample was assessed by gel electrophoresis (1% agarose, ethidium bromide detection) using K562 DNA (250, 100, 50, 25, 12.5, 6 and 3 ng) to estimate the quantity of DNA in each sample.

2.3. nuTH01–nuSRY–nuCSF–IPC qPCR quantifications

The nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay, as optimized for use on the Applied Biosystems 7500 Real Time PCR system, was run in a 20 μL reaction volume that included 10 μL of 2× TaqMan® Universal Master Mix, no UNG (Applied Biosystems), 0.5 μL of 5 U/μL AmpliTaq Gold® enzyme, 1 μL of 3.2 μg/μL non-acetylated BSA (Sigma), and 4 μL of sample. The remaining 4.5 μL in each reaction volume was composed to give final primer, probe and oligonucleotide concentrations as follows: 600 nM each nuTH01 primer; 200 nM nuTH01 probe; 600 nM each nuSRY primer; 200 nM nuSRY probe; 400 nM each nuCSF primer; 100 nM nuCSF probe; 50 nM each IPC primer; 200 nM IPC probe; and 90,000 copies IPC oligonucleotide. Primers (Applied Biosystems for nuCSF and Operon, Huntsville, AL for nuTH01, nuSRY and IPC) were diluted in sterile, de-ionized water. Probes (Operon for nuTH01 and Applied Biosystems for nuSRY, nuCSF and IPC) were diluted in TE^{-4}.

Real-time qPCR data were collected on an Applied Biosystems 7500 SDS instrument controlled by a computer running version 1.3 of the 7500 SDS Collection software. The instrument was configured for the following run conditions: 20 μL sample volumes; 9600 emulation mode; one 10 min 95 °C polymerase activation step, followed by 45 cycles of two-step qPCR (15 s of 95 °C denaturation, 60 s of 60 °C combined anneal/extension). The 45 cycle qPCR parameters were utilized throughout the developmental validation to monitor for contamination in the TE^{-4} negative template controls and to detect cross reactivity in the species studies, but for routine analysis, 40 cycles were used. Well-to-well variations in background fluorescence were corrected for by use of a ROX-labeled passive reference oligonucleotide included as part of the Applied Biosystems TaqMan® Universal Master Mix.

Amplification curves were analyzed with the automatic baseline feature of the 7500 System SDS software and empirically determined thresholds for each portion of the assay (nuTH01 threshold = 0.025; nuSRY threshold = 0.20; nuCSF threshold = 0.10; IPC threshold = 0.025). For each experiment, the 7500 System SDS software used amplification results from a freshly prepared serial dilution series of standard DNA to generate three calibration curves (one for nuTH01, one for nuSRY and one for nuCSF) by plotting C_T (cycle threshold) vs. log C_0 (initial standard DNA concentration). DNA quantifications for unknown samples were interpolated from the resulting linear calibration curves using semi-automatic features of the 7500 System SDS software. For the nuTH01, nuSRY and nuCSF assays, calibration plots were constructed using data from three-fold serial dilution samples containing 32–0.044 ng (duplicate 0.044 ng samples) of Promega human genomic standard DNA. With the exception of the plate precision and population studies, at least one TE^{-4} negative template control was run on each plate. We used the following formula [22] utilizing the slopes of the calibration curves to calculate all assay efficiencies: %PCR efficiency = $(10^{(-1/\text{slope})} - 1) \times 100$. Analyzed data were generally exported from the 7500 SDS software into a CSV file (comma delimited), which was then either opened directly with Microsoft® Excel 2000 or imported into a Microsoft® Excel 2000 spreadsheet for semi-automated reformatting of the data and secondary analysis of the results.

2.4. STR genotyping

The AmpF*l*STR® Identifiler™ and AmpF*l*STR® Yfiler™ PCR Amplification kits (Applied Biosystems) were used for STR genotyping. PCR amplifications of 1 ng of autosomal DNA and 0.5 ng of Y-chromosomal DNA in a 25 μL reaction volume were performed according to vendor instructions on a GeneAmp® 9700 PCR thermocycler (Applied Biosystems). STRs were resolved and detected on an ABI Prism® 310 Genetic Analyzer (Applied Biosystems) according to vendor instructions except that the electrokinetic sample injection voltage was dropped from the default of 15 kV to 5 kV. In addition, 5 and 10-s injections were performed for some experiments. Electrophoresis data were analyzed using GeneMapper™ ID 3.2 (Applied Biosystems) using 50 RFU peak amplitude thresholds and a baseline setting of 35 rather than the default of 51 for the AmpF*l*STR® Identifiler™ data.

3. Results and discussion

In the United States, the Scientific Working Group on DNA Analysis Methods (SWGDAM) establishes guidelines for both

the developmental and internal validation of forensic DNA analysis methods. Our discussion of the developmental validation of the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay follows the organization of the most recent guidelines provided by SWGDAM [19].

3.1. Characterization of genetic markers (SWGDAM 2.1)

Previously, we indicated the criteria for selection of the TH01 and CSF1PO STR loci and the IPC target [17]. Work on the Y target centered on a flanking region (Yp11.3) of the SRY gene, in particular using bases 1726–1965 from GenBank locus L10102 (HUMSRYB) [23]. Several candidate assays were examined at this locus and were compared with respect to Y-chromosome specificity, sensitivity and PCR efficiency (results not shown). Optimal results were obtained using previously described primers [20] in combination with a locally designed TaqManMGB® probe. Sequence information for the nuSRY primers and nuSRY probe are provided in Table 1. As shown in Table 2, several primer and probe concentrations were examined in order to determine the final composition of the quadruplex qPCR assay. The final mix (see bold values in Table 2) provided optimal qPCR efficiencies as well as optimal results for male:female mixtures and hematin-inhibited samples (results not shown).

3.2. Species specificity (SWGDAM 2.2)

Human specificity was addressed by running the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay using 1 and 100 ng of template DNA, in duplicate, from 15 non-human species (sex unknown for higher organisms) and from five known male mammalian samples (cat, cow, dog, horse and pig). At 1 ng template DNA levels the nuSRY, nuTH01 and nuCSF assays showed no evidence of cross-reactivity to DNA from any of the organisms tested and at 100 ng template DNA levels the known male mammalian samples showed no evidence of cross-reactivity to the nuSRY, nuTH01 and nuCSF assays. Of the 100 ng amplifications, only the monkey sample yielded a reproducible signal, a nuCSF average value of 28 pg, indicating a sensitivity ~3500 times lower than for human DNA. Non-reproducible, low-level signal (<23 pg) were seen in one each of the two duplicates for the cat (nuCSF only), mouse (nuCSF only), *B. subtilis* (nuSRY only) and *C. perfringens* (nuSRY and nuTH01). Consistent with the lack of any observable cross-reactivity of the nuSRY assay to non-human, male, mammalian samples, BLASTn [v. 2.2.15 October-15-2006] [24] searches for homologies to the nuSRY target sequence revealed significant similarities only to SRY sequences for three primate species (chimpanzee, gorilla and vervet monkey). Neither BLASTn nor BLAT [Human Build Mar. 2006 May-14-2007] [25] analysis revealed homologous sequences for any non-primate species that would be expected to be amplified by the nuSRY primers. As previously indicated, specificity studies at the TH01 and CSF1PO STR loci indicate that the nuTH01 and nuCSF assays will detect DNA from non-human higher primates [26,27].

Table 2
Primer and probe concentrations utilized in nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay development

nuTH01 primer concentration (nM)	nuTH01 probe concentration (nM)	nuSRY primer concentration (nM)	nuSRY probe concentration (nM)	nuCSF primer concentration (nM)	nuCSF probe concentration (nM)	IPC primer concentration (nM)	IPC probe concentration (nM)	IPC oligonucleotide copies	nuTH01 %PCR efficiency[a]	nuSRY %PCR efficiency[a]	nuCSF %PCR efficiency[a]
600	*200*	*N/A*	*N/A*	*400*	*100*	*100*	*100*	*90,000*	*82*	*N/A*	*86*
600	200	300	100	400	100	200	100	90,000	86	94	96
600	200	300	100	400	100	100	100	90,000	87	91	88
600	200	300	200	400	100	50	200	90,000	87	90	83
600	**200**	**600**	**200**	**400**	**100**	**50**	**200**	**90,000**	93	93	92
600	200	400	200	400	100	100	200	90,000	96	86	91
600	200	400	200	400	100	75	200	90,000	98	93	90
600	200	400	200	400	100	100	200	90,000	92	87	86
600	200	500	200	400	100	100	200	90,000	97	87	86

Italicized values indicate concentrations and results for the nuTH01–nuCSF–IPC triplex qPCR assay. Bold numerals indicate the optimal concentrations used for the final version of the quadruplex assay.
[a] %PCR efficiency = $(10^{-1/\text{slope from the standard DNA dilution series}} - 1) \times 100$.

3.3. Sensitivity (SWGDAM 2.3)

To determine the range of input DNA able to produce reliable quantification results, we completed a sensitivity study using replicate samples. We tested the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay on a 100 ng–7.2 pg dilution series of Human Genomic Male DNA (Promega) using the stock concentration provided by the vendor as the known starting concentration. Except for the IPC assay, no amplification signals were evident in the negative control samples (TE^{-4}).

The nuTH01, nuSRY and nuCSF targets showed acceptable precision, as indicated by standard deviations in C_T less than one cycle, down to ~44 pg of template. At a template level of 7.2 pg, the standard deviation in C_T was greater than one cycle at the nuTH01 target and the nuSRY target showed loss of amplification (''signal dropout'') in 20% of the replicates as did 40% of the replicates with the nuCSF assay. Also, the calibration curves for the nuTH01, nuSRY and nuCSF assays showed good linearity ($R^2 = 0.99$) using qPCR data over the entire template range of 100 ng–7.2 pg (Fig. 1) with C_T values less than 40. The assay efficiencies were 92%, 95% and 98% for the nuTH01, nuSRY and nuCSF assays, respectively.

The IPC assay amplifies an exogenous template oligonucleotide that is contained within the primer/probe mix and is therefore added to each sample in equivalent quantities. As a result, the same number of IPC template copies is available for amplification in each reaction regardless of the quantity of genomic DNA. Consequently, the IPC C_T values are expected to remain fairly constant from sample to sample unless the PCR efficiency for a sample is reduced due to the presence of PCR inhibitors or the presence of excessive quantities of human DNA. For samples with 10 ng or less of standard DNA, the mean IPC C_T was 25.16 with a standard deviation of 0.24 cycles. When the standard DNA quantity exceeded 10 ng (the 32 and 100 ng samples), significant delays in the IPC amplification were seen. This phenomenon probably occurred as a result of the genomic DNA amplifications out-competing the primer-limited IPC amplification and reducing the efficiency of the IPC PCR [28,29]. If these samples with higher template amounts are included, the mean C_T for the IPC was 25.59 with a standard deviation of 1.28 cycles. These results indicate that care should be taken when interpreting the IPC C_T for samples with more than 10 ng of DNA; for such samples, delayed IPC C_T values could be due to high template amounts and/or PCR inhibition.

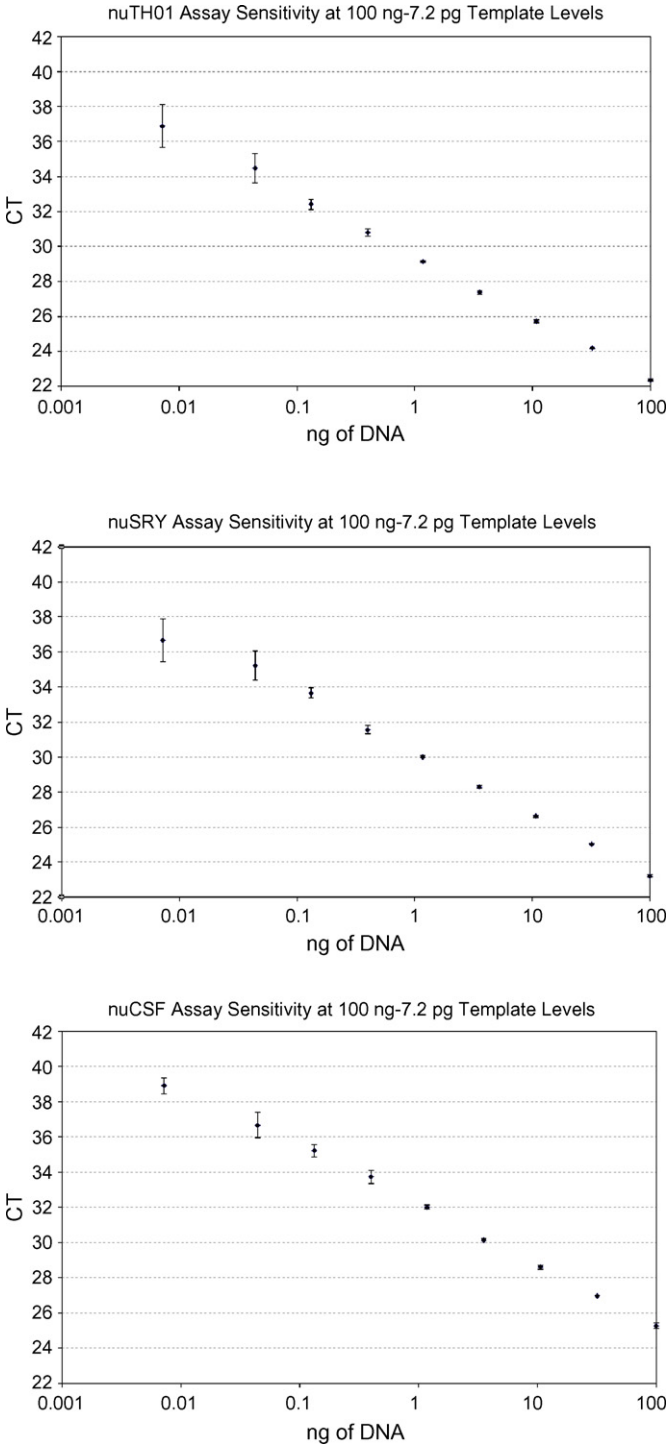


Fig. 1. Sensitivity of nuTH01–nuSRY–nuCSF–IPC assay. Three-fold serial dilutions of Promega Human Genomic Male DNA used to create standard curves from all reactions containing 100 ng–7.2 pg total template. *Note*: Data for detected standards only; *y*-axis error bars represent one standard deviation.

3.4. Stability (SWGDAM 2.4)

Because the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay is expected to be useful for samples that are degraded and/or inhibited, it was imperative to determine its ability to provide accurate results from samples that represented this type of environmental abuse. As a result, two separate studies, one using DNase I-degraded DNA to simulate environmentally degraded DNA and one using DNA spiked with hematin to simulate DNA co-extracted with PCR inhibitors, were conducted to satisfy this guideline.

We first evaluated the ability of the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay to detect DNA degradation and thereby to predict the success of downstream STR analysis. This assessment was made by performing time-course exposure of high molecular weight female (HL60) DNA and

male (Raji) DNA to the DNA nuclease DNase I to generate a series of degraded female DNA and a series of degraded male DNA with varying levels of degradation. The time-points used to create the degradation series varied from 0 to 92 min with varying concentrations of DNase I as previously indicated [17]. DNase-treated samples at each of the time-points were first examined by agarose gel electrophoresis to determine the average size of the DNA fragments at each selected time-point. Final series were constructed by selecting six DNase-treated male and six DNase-treated female samples with varying degrees of degradation in addition to a non-degraded sample (Supplementary Figure 1). These samples were diluted to 1 ng/μL relative to the starting concentrations of the stock DNA as indicated on the manufacturers labeling. Mixtures of these degraded male and degraded female DNA samples were prepared at 1:1 and 1:10 ratios by mixing lightly degraded DNAs M1 and F1 (majority of fragments >2027 bp), mildly degraded DNAs M2 and F2 (majority of fragments between 564 and 2027 bp) and highly degraded DNAs M6 and F6 (majority of fragments <564 bp). The degraded samples and mixtures of degraded DNA were then tested with the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay to determine the quantity of amplifiable DNA and the extent of degradation present in each time-point.

The detected quantities of DNA decreased and the qPCR degradation ratios (nuCSF quantity/nuTH01 quantity) generally increased as DNase exposure time and/or the ratio of DNase to DNA increased (Table 3). The one exception was the most highly degraded female DNA sample, F6, that yielded a qPCR degradation ratio that was lower than expected. As expected, because of their dissimilar amplicon lengths the nuTH01, nuSRY and nuCSF portions of the assay experienced different sensitivities to the level of degradation in the DNase-treated samples. The qPCR-based degradation ratios generally displayed a good progression with respect to increasing DNase treatment time as did the samples on the agarose gel (Supplementary Figure 1).

The main objective of accurately quantifying nuclear DNA in a forensic sample is to ensure that the appropriate quantity of template DNA is used in the multiplexed STR amplification. Whenever possible, 0.5 ng of each DNase-degraded male sample, as determined by the nuSRY quantification, was amplified with the AmpF/STR® Yfiler™ kit to assess the assay's ability to properly quantitate input template for Y STR amplifications. Likewise, 1 ng of each DNase-degraded sample, as determined by the nuTH01 quantification, was amplified with the AmpF/STR® Identifiler™ kit in to assess the assay's ability to properly quantitate input template for autosomal STR amplifications. The resulting STRs were resolved and detected on the ABI 310 and the percent of detected alleles was determined. Across the set of DNase-degraded samples, an increase in the qPCR degradation ratio and in interlocus imbalance in the STR profiles correlated with greater degradation. Each of the non-mixed female and male DNA samples treated with DNase had 100% of the peaks detected with the standard 5-s injections (Table 3 and Fig. 2a–c). No off-scale peaks or pull-up peaks were detected in any of the AmpF/STR®

Table 3
AmpF/STR® Identifiler™ and AmpF/STR® Yfiler™ STR results for DNase-degraded male and female DNA

Male DNA (Raji)								Female DNA (HL60)				
Sample	Mean qPCR degradation ratio[a]	D19/D18 STR ratio[b]	Average TH01 peak height (RFU)	Identifiler™ peaks detected (%)	DYS393/DYS392 STR ratio[b]	DYS393 peak height (RFU)	Yfiler™ peaks detected (%)	Sample	Mean qPCR degradation ratio[a]	D19/D18 STR ratio[b]	Average TH01 peak height (RFU)	Identifiler™ peaks detected (%)
M0	1.1	1.1	1038	100	1.7	713	100	F0	1.3	1.2	2032	100
M1	1.3	1.9	1184	100	1.4	744	100	F1	1.7	1.2	1145	100
M2	1.8	4.1	1215	100	2.3	563	100	F2	2.3	1.6	1215	100
M3	3.0	7.1	1221	100	2.2	422	100	F3	2.8	2.0	1349	100
M4	4.5	6.5	1038	100	3.3	403	100	F4	5.8	3.3	721	100
M5	6.3	11	989	100	3.7	408	100	F5	8.4	4.4	847	100
M6	8.2	17	793	100	5.6	600	100	F6	5.4[c]	5.4	742	100

Note: Samples designated M0–M6 and F0–F6 represent DNase-degraded male and female DNA with increasing level of degradation, respectively.

[a] qPCR degradation ratio = nuCSF quantity/nuTH01 quantity.

[b] STR ratio = peak height of the smallest STR locus/peak height of largest STR locus.

[c] The qPCR degradation ratio is lower than expected for these samples, but is in agreement with the D19/D18 STR ratio.

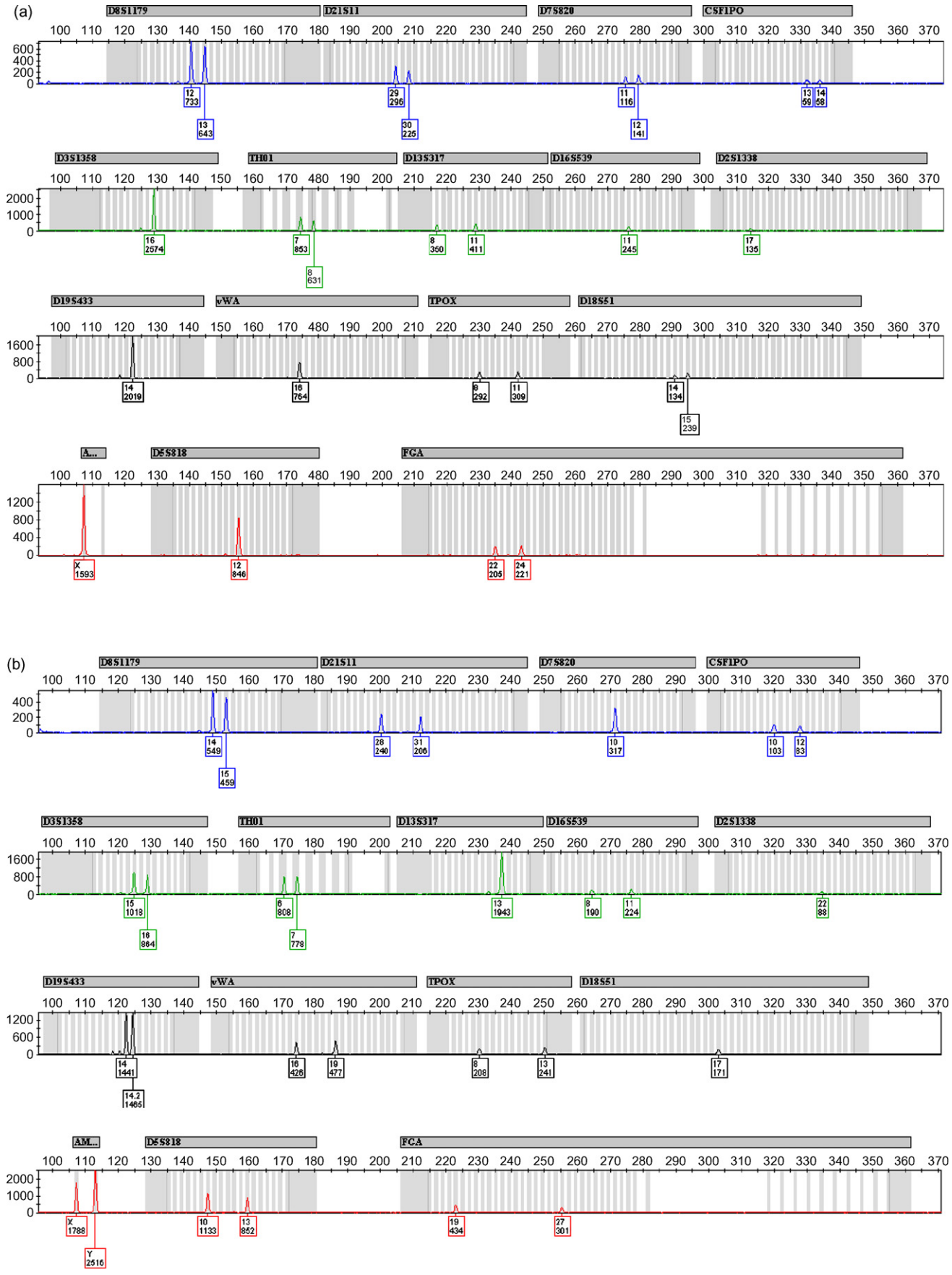


Fig. 2. (a) AmpF*l*STR® Identifiler™ STRs from DNase-degraded female (HL60) DNA (Sample F6). (b) AmpF*l*STR® Identifiler™ STRs from DNase-degraded male (Raji) DNA (Sample M6). (c) AmpF*l*STR® Yfiler™ STRs from DNase-degraded male (Raji) DNA (Sample M6).

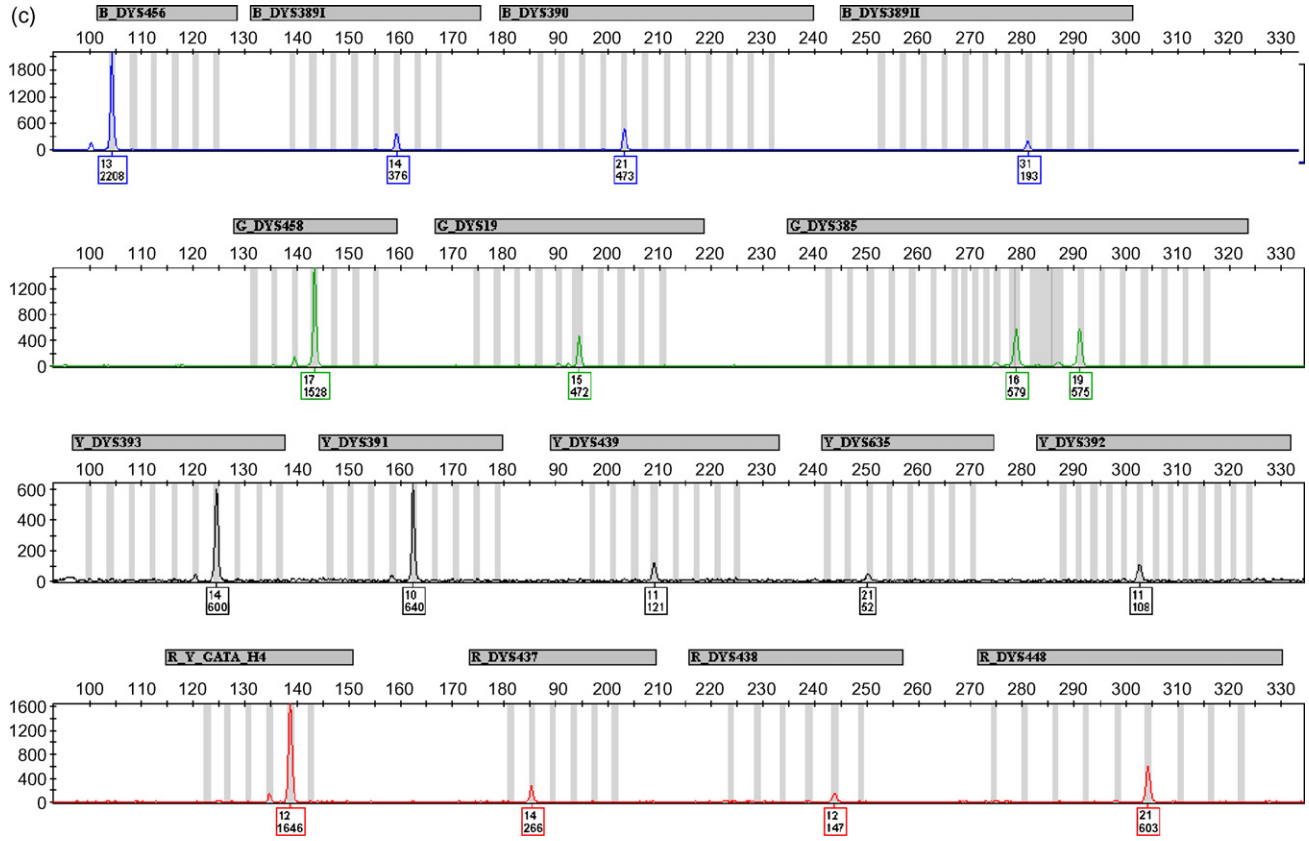


Fig. 2. (*Continued*).

Identifiler™ or AmpF*l*STR® Yfiler™ injections. Mixtures of degraded male and female DNA quantified with the quadruplex assay yielded expected AmpF*l*STR® Yfiler™ and AmpF*l*STR® Identifiler™ typing results that were consistent with the quantities amplified and the known mixture ratios (Supplementary Table 1).

The results in Table 3 indicate that the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay can determine appropriate template DNA quantities for the amplification and detection of Y and autosomal STRs in degraded samples. In particular, for both sets of degraded DNA samples, average autosomal TH01 STR peak heights were readily interpretable, falling in the range of 721–1349 RFUs. For the male set of degraded DNA samples, the DYS393 Y STR peak heights were also readily interpretable (403–744 RFUs). Moreover, despite the increasing degree of DNA fragmentation in these samples, reflected by both the increasing qPCR degradation ratios and the increasing STR ratios listed in Table 3, 100% of the autosomal and Y STR peaks reached the 50 RFU interpretation threshold. The effect of DNA degradation on STR-amplified samples was estimated by comparing the peak heights of selected short and long STR loci. For the AmpF*l*STR® Identifiler™ kit, degradation was estimated for each sample by calculating the ratio of the peak heights of the short D19S4433 locus to the long D18S51 locus. For the AmpF*l*STR® Yfiler™ kit, degradation was estimated by the peak–height ratio of the short DYS392 locus to the long DY393 locus. These ratios showed similar trends to the qPCR degradation ratio, and to the increasing degree of fragmentation observed on the agarose gel (Table 3 and Supplementary Figure 1).

We evaluated the sensitivity of the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay to inhibition by amplifying samples containing various concentrations of hematin. The levels of inhibition in these samples were also examined with the AmpF*l*STR® Identifiler™ and AmpF*l*STR® Yfiler™ STR amplification kits. Eleven DNA samples with hematin concentrations in the qPCR reactions of 0, 10, 20, 30, 40, 50, 60, 70, 80, 120 and 160 μM were quantified in duplicate.

Each of the targets (nuTH01, nuSRY, nuCSF and IPC) in the quadruplex qPCR assay eventually showed delayed C_Ts with increasing hematin concentration, relative to those measured for non-inhibited samples (Fig. 3 and Table 4). Even though null IPC amplifications were only observed at the highest hematin concentration (160 μM) other samples with lower concentrations of hematin showed delayed IPC C_T values indicative of substantial PCR inhibition, which resulted in a significant underestimation of the nuclear DNA (Table 4). IPC C_T delays were determined by subtracting the mean C_T for a set of samples *known* to be free of PCR inhibitors (DNA standard samples < 15 ng) from the IPC C_T of the hematin-spiked sample. We currently use this IPC C_T delay, labeled as the delta IPC (or ΔIPC), to identify samples that may contain significant quantities of PCR inhibitors. A ΔIPC value greater than 0.75 may indicate an inaccurate DNA quantification due to PCR inhibition [17], which may require re-quantification after

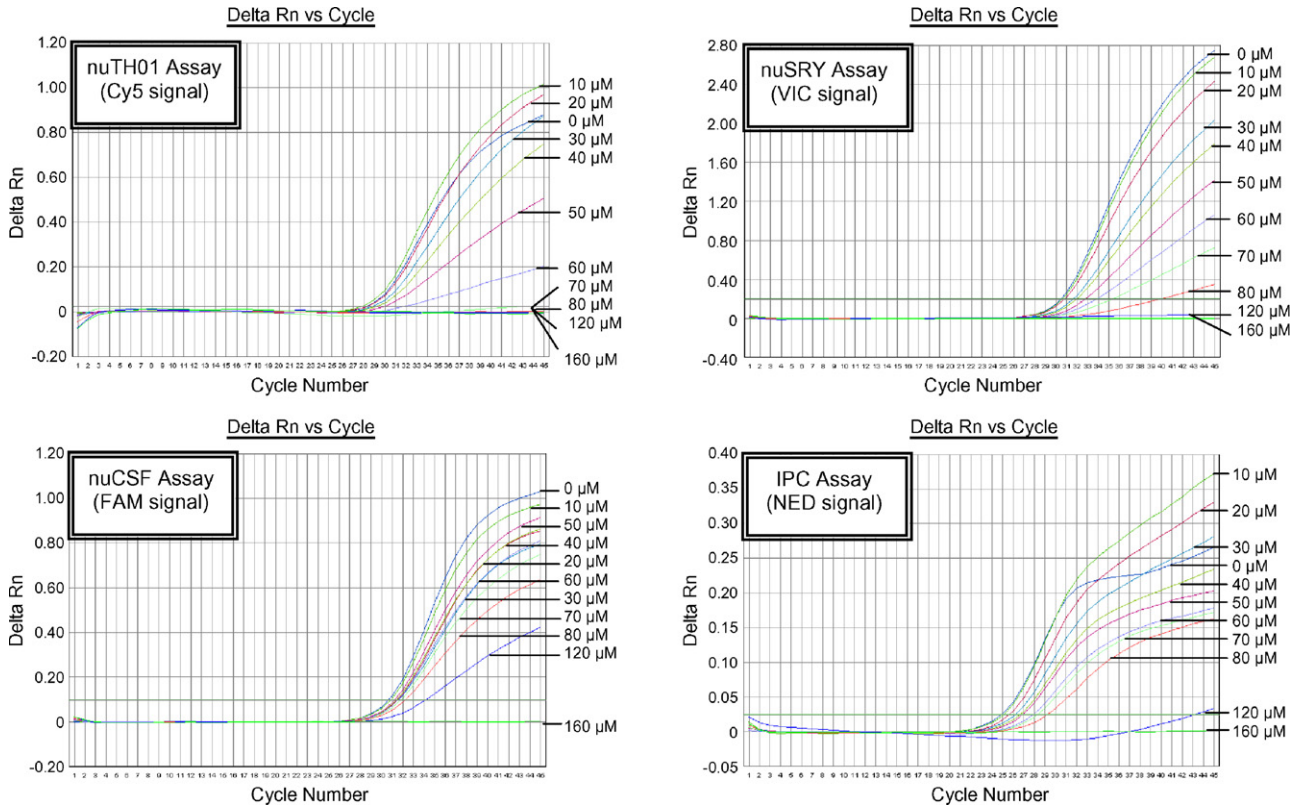


Fig. 3. Amplification curves for hematin-inhibited samples. Although all samples were assayed in duplicate, for clarity only a single replicate is displayed for each. Concentrations indicate final concentration of hematin in the qPCR amplification.

sample dilution or sample clean-up to avoid amplification of excess template in the STR reaction.

It should be noted that the ΔIPC value should not be used by itself to predict the success or failure of the subsequent STR amplification as it does not take into consideration the quantity of DNA and/or PCR inhibitors in the STR reaction relative to the quantity in the qPCR reaction. We suggest here a simple and general approach to address this issue. For each sample quantified by qPCR, a ''normalized inhibition factor'' (NIF) is calculated to estimate the degree of inhibition per nanogram of sample:

$$\text{NIF} = \frac{\Delta\text{IPC (cycles)}}{\text{quantity of DNA (ng)}}$$

The larger the NIF, the more likely the sample is to exhibit PCR inhibition effects in downstream STR amplifications. Although we will discuss and evaluate the NIF values in the context of our nuTH01–nuSRY–nuCSF–IPC quadruplex assay, the approach could be used, in principle, for any qPCR assay that includes an IPC.

For the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay, it is possible to calculate three independent NIF values for each sample: a nuTH01 autosomal NIF (ΔIPC/ng nuTH01), a nuCSF autosomal NIF (ΔIPC/ng nuCSF) and a nuSRY Y-chromosomal NIF (ΔIPC/ng nuSRY). As shown in Table 5, the nuTH01 autosomal NIF is useful for assessing potential inhibition effects in downstream Identifiler™ STR amplifications, while the nuSRY Y-chromosomal NIF is useful for assessing similar effects in downstream Yfiler™ STR amplifications. In particular, samples with nuTH01 autosomal NIF values greater than ∼1 are seen to produce incomplete Identifiler™ profiles or null amplifications. These results effectively establish a *threshold* of ∼1 for the nuTH01 autosomal NIF, which, if reached, is indicative of *potential* PCR inhibition in downstream Identifiler™ STR amplifications. Table 5 also shows that samples with nuSRY Y-chromosomal NIF of ∼2 are likely to produce incomplete Yfiler™ profiles or null amplifications, effectively establishing a Y-chromosomal NIF threshold of ∼2 for predicting downstream amplification effects. It is possible that the increase in the NIF threshold value from ∼1 for the autosomal nuTH01 value to ∼2 for the Y-chromosomal nuSRY value is a reflection of the different template amounts used in the two STR assays; twice as much DNA (and PCR inhibitor) is utilized in the 1 ng Identifiler™ reactions than in the 0.5 ng Yfiler™ reactions, and thus the ratio of PCR inhibitor to DNA can be twice as high before it has an equivalent effect. Although we do not currently use nuCSF autosomal NIF values, these might be helpful for predicting downstream inhibition effects in MiniSTR amplifications. Finally, although our results indicate that NIF values are potentially useful for identifying samples that may show inhibition in downstream PCR amplifications, the NIF approach does not cover every situation. For example, some samples that contain low quantities of DNA *and* PCR inhibitors may yield insignificantly small ΔIPC values, especially when a

Table 4
Hematin inhibition of quadruplex qPCR, AmpF/STR® Identifiler™ and AmpF/STR® Yfiler™ STRs

nuTH01–nuSRY–nuCSF–IPC qPCR						Identifiler™			Yfiler™		
Hematin in qPCR assay (μM)	Mean IPC C_T (S.D.) (cycles)	ΔIPC[a] (cycles)	nuTH01 (ng/μL)	nuSRY (ng/μL)	Mean qPCR degradation ratio[b]	Volume of template in STR amplification (μL)	Hematin in STR amplification (μM)	Peaks detected for 5-s injection (%)	Volume of template in STR amplification (μL)	Hematin in STR amplification (μM)	Peaks detected for 5-s injection (%)
0	24.98 (0.08)	−0.18	0.57	0.47	1.2	1.75	0	100	1.08	0	100
10	25.05 (0.13)	−0.10	0.66	0.42	0.9	1.52	3	100	1.20	2	100
20	25.65 (0.01)	0.50	0.54	0.34	1.6	1.84	7	100	1.49	6	100
30	26.21 (0.03)	1.1	0.35	0.23	1.1	2.84	17	100	2.14	13	100
40	26.63 (0.06)	1.5	0.30	0.17	1.6	3.36	27	39	2.93	23	76
50	27.04 (0.14)	1.9	0.17	0.10	2.8	5.74	57	0	4.87	49	0
60	27.82 (0.15)	2.7	0.06	0.05	6.5	10.00	120	0	10.00	120	0
70	28.21 (0.10)	3.1	–	0.02	15[c]	10.00	140	0	10.00	140	0
80	29.18 (0.14)	4.0	–	0.002	–	10.00	160	0	10.00	160	0
120	42.19 (1.47)	17	–	–	–	10.00	240	0	10.00	240	0
160	–	–	–	–	–	10.00	320	0	10.00	320	0

[a] Delays in IPC C_T (ΔIPC) were determined by subtracting the mean C_T for a set of samples *known* to be free of PCR inhibitors (DNA standard samples < 15 ng) from the IPC C_T of the hematin-spiked sample.
[b] qPCR degradation ratio = nuCSF quantity/nuTH01 quantity.
[c] Value from single measurement as only one replicate yielded a nuTH01 value greater than our 12.5 pg limit of quantitation.

small volume (1–4 μL) of sample is amplified with the nuTH01–nuSRY–nuCSF–IPC quadruplex assay. For such samples, the larger quantities and volumes utilized in the STR amplification may contain sufficient PCR inhibitor(s) to adversely impact amplification. Finally it should be noted that the NIF thresholds described above were determined using a diffusible inhibitor (hematin) model, and this model may not reflect every type of PCR inhibition encountered.

For the set of hematin-inhibited samples, 1 ng of template was used in each AmpF/STR® Identifiler™ amplification and 0.5 ng of template was used in each AmpF/STR® Yfiler™ amplification based on concentrations determined from the nuTH01 and nuSRY portions of the qPCR assay. If, based on the nuSRY or nuTH01 qPCR assays, the use of 0.5 or 1 ng of template was not possible, 10 μL of the sample was used. The hematin concentration ranged from 0 to 320 μM in both the 25 μL AmpF/STR® Identifiler™ and AmpF/STR® Yfiler™ reactions.

For the AmpF/STR® Identifiler™ reactions, each sample was injected at 5 kV for 5 and 10 s on the ABI 310 Genetic Analyzer and analyzed with GeneMapper 3.2 with a 50 RFU analysis threshold. Complete genotypes were obtained with samples that had a final hematin concentration of 17 μM or less with autosomal NIF values less than 1 (Tables 4 and 5 and Fig. 4a). Only 39% of the peaks were detected at the 27 μM hematin concentration when a 5-s injection was used (Table 4). Increasing the injection time from 5 to 10 s increased the percentage of peaks detected in this sample from 39% to 46% (Table 5). The AmpF/STR® Identifiler™ amplification was completely inhibited at hematin concentrations of 57 μM and autosomal NIF values significantly greater than 1 (Table 5 and Fig. 4a). Increasing the injection time from 5 to 10 s did not improve allele detection in these samples. The results from the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay provided an indication of which samples would fail STR analysis because of the presence of a PCR inhibitor as indicated by autosomal NIF values greater than 1 (Table 5 and Fig. 4a).

For the AmpF/STR® Yfiler™ reactions, each sample was injected at 5 kV for 5 and 10 s on the ABI 310 Genetic Analyzer and analyzed with GeneMapper 3.2 with a 50 RFU analysis threshold. Complete genotypes were obtained with samples that had a final hematin concentration of 13 μM or less with Y-chromosomal NIF values less than 1 (Tables 4 and 5 and Fig. 4b). Only 76% of the peaks were detected at 23 μM hematin concentration when a 5-s injection time was used (Table 4). Increasing the injection time from 5 to 10 s enhanced the percentage of peaks detected from 76% to 88% (Table 5). The AmpF/STR® Yfiler™ amplification was completely inhibited at hematin concentrations of 49 μM or above and Y-chromosomal NIF values significantly greater than 2 (Table 5 and Fig. 4b). Increasing the injection time from 5 to 10 s did not improve allele detection in these samples. The results from the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay provided an indication of which samples would fail STR analysis because of the presence of a PCR inhibitor as indicated by the Y-chromosomal NIF values greater than 2 (Table 5 and Fig. 4b).

Table 5
Hematin inhibition and the normalized inhibition factor (NIF) for AmpF*l*STR® Identifiler™ and Yfiler™ STR typing[a]

Hematin (μM)	ΔIPC (cycles)[a]	nuTH01 (ng)[b]	Autosomal NIF normalized inhibition factor (ΔIPC/nuTH01 (ng))	Identifiler™ peaks detected (%)	nuSRY (ng)[c]	Y-chromosomal NIF normalized inhibition factor (ΔIPC/nuSRY (ng))	Yfiler™ peaks detected (%)
0	−0.18	2.3	−0.080	100	1.9	−0.090	100
10	−0.10	2.7	−0.040	100	1.7	−0.060	100
20	0.50	2.2	0.23	100	1.4	0.37	100
30	1.1	1.4	0.79	100	0.92	1.2	100
40	1.5	1.2	1.3	39[d]/46[e]	0.68	2.2	76[d]/88[e]
50	1.9	0.7	2.7	0	0.40	4.8	0
60	2.7	0.24	11	0	0.20	14	0
70	3.1	0.08	39	0	0.08	39	0
80	4.0	0.00	–	0	0.01	500	0
120	17	0.00	–	0	0.00	–	0
160	–	–	–	0	–	–	0

[a] Values from Table 4. Delays in IPC C_T (ΔIPC) were determined by subtracting the mean C_T for a set of samples *known* to be free of PCR inhibitors (DNA standard samples < 15 ng) from the IPC C_T of the hematin-spiked sample.
[b] *Apparent* DNA quantity determined from nuTH01 portion of quadruplex qPCR assay.
[c] *Apparent* DNA quantity determined from nuSRY portion of quadruplex qPCR assay.
[d] Data from 5-s injection.
[e] Data from 10-s injection.

3.5. Reproducibility (SWGDAM 2.5)

Nineteen samples (12 extracts from quality control bloodstains and 7 purchased DNA control samples) measured with the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay in two independent qPCR runs were compared to assess reproducibility. The two runs were performed by different analysts who used the same reagents and the same stock solutions of sample DNA for each run, but who independently prepared fresh dilutions for the standard curves using Human Genomic Male DNA (32 ng–44 pg standard dilution range). In both runs, each sample was quantified once.

The two runs were analyzed separately, but with identical analysis parameters in order to determine the inter-run reproducibility. The inter-run data for each sample was used to calculate the mean, standard deviation and percent coefficient of variation of the nuTH01, nuSRY and nuCSF quantifications (Table 6). The inter-run precision in quantitation, expressed as the percent coefficient of variation (%CV = 100 × (standard deviation/mean)), had an average of 4.0% with an 11% maximum for the nuTH01 portion of the assay. For the nuSRY portion of the assay, the inter-run precision in quantitation had an average of 9.5% with a 29% maximum. For the nuCSF portion of the assay, the inter-run precision in quantitation had an average of 12% with a 25% maximum. The inter-run precision in the qPCR degradation ratio had an average of 15% with a 35% maximum and the inter-run male:female (nuSRY:nuTH01) ratio had an average of 12% with a 36% maximum. The run-to-run reproducibility for the IPC was determined by comparing the C_T values obtained for each sample. The inter-run standard deviations in C_T were consistently less than 0.3 cycle; however, because the IPC amplification is dependent on each analyst's preparation of the qPCR master mix, the intra-run precision in the IPC C_T is more meaningful. The standard deviation in C_T, within a run, was 0.14 cycles (data not shown).

3.6. Case-type samples (SWGDAM 2.6)

The goal of this study was to determine the ability of the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay to accurately quantitate DNA in a variety of known and non-probative samples. This assessment was made by quantifying 48 case-type samples (not counting multiple dilutions of each sample), then using the resulting nuSRY and nuTH01 quantification values to perform AmpF*l*STR® Yfiler™ STR and AmpF*l*STR® Identifiler™ STR amplifications, respectively.

Each sample containing male DNA, as indicated by the nuSRY assay, was amplified with the AmpF*l*STR® Yfiler™ kit (0.5 ng/10 μL) with two exceptions (Table 7). The first exception was sample 79 (non-sperm fraction of post-coital vaginal swab from a vasectomized male/female coupling) where only ~300 pg of male DNA was contained in a 10 μL extract that also contained ~2000 ng of female DNA. The second exception was sample 82 (male sweat mixed with female saliva) where only ~4 pg of male DNA was detected in the 4 μL sample and ~10 pg of male DNA was contained in the 10 μL extract. Twenty-two of the 24 samples amplified with the AmpF*l*STR® Yfiler™ kit yielded full profiles. As expected, no alleles were detected in sample 82, which is consistent with the low quantity of template amplified. Also, sample 79 did not yield a full profile (Fig. 5) with only the DYS392 locus/allele not reaching threshold. This result is not unexpected, as less than 500 pg of male DNA was amplified for this sample. The STR results indicate that the quadruplexed nuSRY target effectively estimates the quantity of male DNA in many sample types (Table 7).

The nuTH01 target was used in previous duplex and triplex qPCR assays [17,18,23] and was shown to adequately quantify DNA for STR typing in a wide variety of case-type samples. To assess the performance of the nuTH01 target in the current quadruplex assay, each case-type sample that had not been

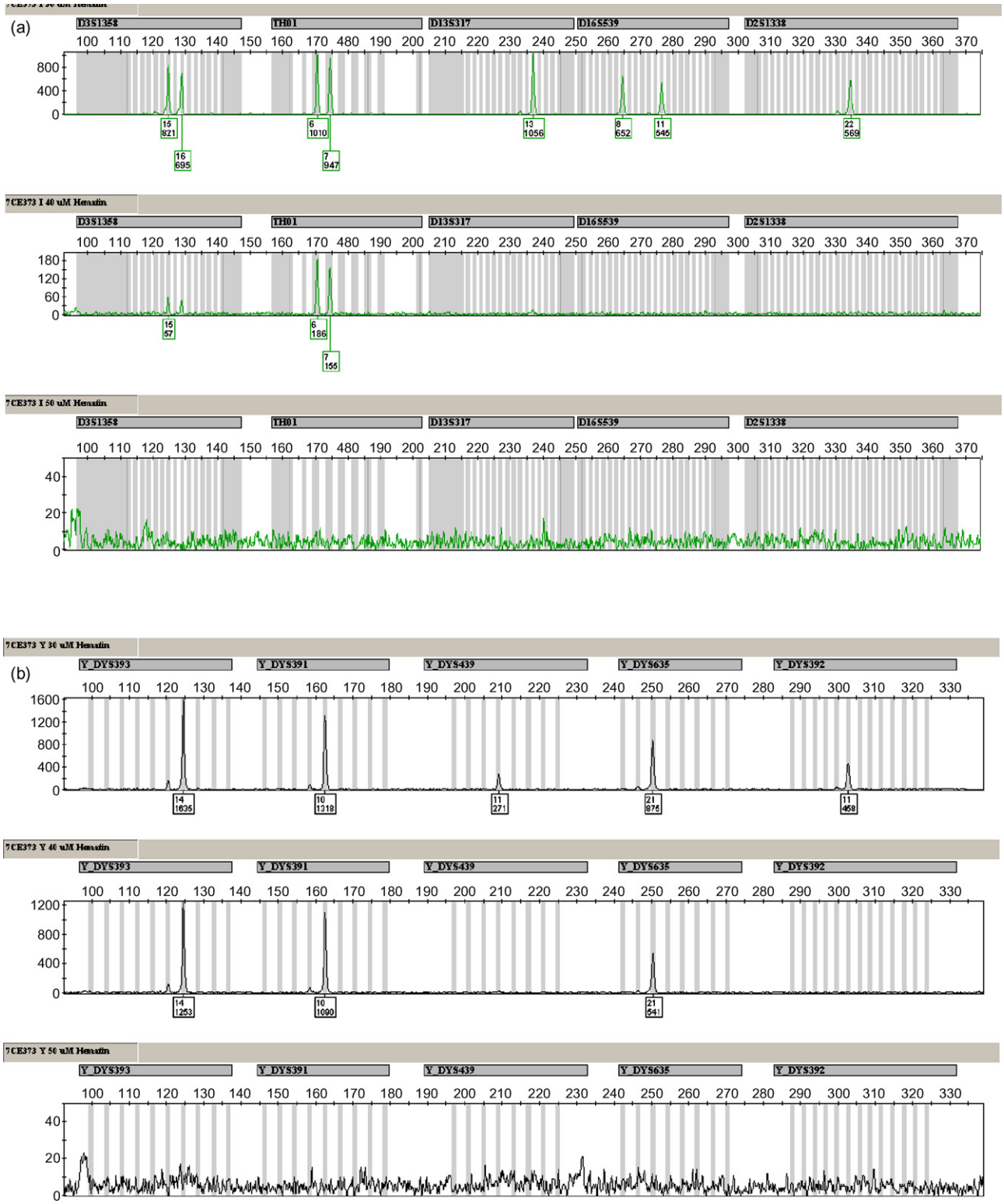


Fig. 4. (a) AmpF*l*STR® Identifiler™ VIC-labeled STRs from hematin-inhibited male (Raji) DNA 17, 27 and 57 μM hematin concentrations from top to bottom with autosomal NIF values of 0.79, 1.3 and 2.7, respectively. (b) AmpF*l*STR® Yfiler™ NED-labeled STRs from hematin-inhibited male (Raji) DNA 13, 23 and 49 μM hematin concentrations from top to bottom with Y-chromosomal NIF values of 1.2, 2.2 and 4.8, respectively.

consumed in the qPCR and/or Yfiler amplifications, was amplified with the AmpF*l*STR® Identifiler™ kit. All 36 of the amplifications containing the optimal quantity of template (1 ng/10 μL) and 6 of 8 amplifications containing suboptimal quantities of template (~0.07–0.23 ng/10 μL) yielded full profiles. As expected, the amplification of less-than-optimal quantities of template generally yielded lower nuTH01 peak heights and amplification of less than 0.07 ng of template

Table 6
Reproducibility of nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay

Sample	nuTH01–nuCSF–IPC triplex qPCR		nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR							
	nuTH01 quantity (ng/μL)	nuCSF quantity (ng/μL)	nuTH01 quantity (ng/μL)[a,e]	Difference from triplex[b] (%)	nuCSF quantity (ng/μL)[a,e]	Difference from triplex[b] (%)	qPCR degradation ratio[c]	nuSRY quantity (ng/μL)	Male:human ratio[d] nuSRY: nuTH01	IPC C_T cycles[e]
9947A	0.08	0.13	0.07 (0.81)	−18	0.11 (22)	−9.4	0.59 (23)	N/A	N/A	25.18 (0.11)
QuantiBlot™ Std A	0.90	1.09	0.90 (0.98)	0.42	1.13 (8.9)	4.5	0.80 (9.9)	N/A	N/A	25.07 (0.05)
HL60	0.44	0.49	0.44 (2.8)	0.28	0.47 (14)	−4.6	0.96 (17)	N/A	N/A	25.34 (0.04)
Quantifiler™ Std	0.43	0.44	0.43 (7.5)	−0.58	0.39 (11)	−11	1.11 (18)	0.35 (24)	0.82 (31)	25.22 (0.23)
Promega Male	1.87	1.85	1.80 (1.9)	−3.4	1.68 (3.5)	−9.0	1.07 (5.3)	1.8 (5.6)	1.00 (7.4)	25.65 (0.09)
Promega Female	1.18	1.25	1.31 (5.5)	11	1.16 (4.3)	−7.0	1.13 (9.8)	N/A	N/A	25.29 (0.17)
TaqMan® Std	0.33	0.34	0.34 (1.6)	5.0	0.27 (19)	−21	1.28 (18)	0.32 (1.7)	0.93 (0.12)	25.28 (0.28)
QC 11	0.74	0.95	0.80 (1.1)	9.4	0.91 (11)	−4.0	0.89 (12)	0.75 (5.4)	0.93 (6.5)	25.34 (0.05)
QC 19	0.45	0.55	0.43 (1.2)	−2.5	0.56 (10)	0.91	0.79 (9.0)	0.39 (8.2)	0.90 (6.9)	25.24 (0.15)
QC 42	1.63	1.83	1.52 (0.23)	−6.9	2.13 (2.3)	16	0.71 (2.1)	N/A	N/A	25.21 (0.11)
QC 62	0.95	1.10	0.86 (2.3)	−9.4	1.42 (3.5)	29	0.60 (1.2)	0.91 (12)	1.06 (10)	25.38 (0.08)
QC 77	0.58	0.79	0.53 (1.0)	−7.6	0.96 (11)	21	0.56 (9.5)	0.55 (3.9)	1.04 (4.9)	25.19 (0.17)
QC 4394	0.91	0.93	0.86 (6.8)	−5.6	0.94 (18)	0.94	0.92 (11)	N/A	N/A	25.13 (0.01)
QC 3488	0.20	0.28	0.20 (9.4)	−3.9	0.30 (15)	7.7	0.68 (24)	0.19 (2.8)	0.98 (12)	25.09 (0.16)
QC 3543	0.73	0.96	0.65 (1.6)	−11	0.94 (10)	−2.5	0.70 (8.7)	0.78 (2.0)	1.20 (0.41)	25.33 (0.02)
QC 3576	0.23	0.29	0.18 (7.2)	−20	0.28 (21)	−4.9	0.68 (28)	0.20 (29)	1.10 (36)	25.19 (0.20)
QC 6140	0.26	0.26	0.23 (11)	−12	0.20 (13)	−23	1.17 (23)	N/A	N/A	25.23 (0.13)
QC 6160	0.27	0.29	0.25 (2.3)	−10	0.27 (11)	−7.1	0.92 (13)	N/A	N/A	25.29 (0.06)
QC 6235	0.39	0.29	0.36 (11)	−7.1	0.32 (25)	11	1.18 (35)	N/A	N/A	25.26 (0.19)

N/A: not applicable to this sample as this was female sample with no nuSRY value for comparison.

[a] Percent coefficient of variation = (standard deviation/mean) × 100.

[b] Difference from triplex (%) = ((quadruplex quantity − triplex quantity)/triplex quantity) × 100.

[c] qPCR degradation ratio = nuCSF quantity/nuTH01 quantity.

[d] Male:human ratio = nuSRY quantity/nuTH01 quantity.

[e] Standard deviation or percent coefficients of variation, where appropriate, in parenthesis.

Table 7
Summary of nuTH01–nuSRY–nuCSF–IPC qPCR quantification, AmpF/STR® Identifiler™ and AmpF/STR® Yfiler™ STR results for selected casework-type samples

Sample no.	Sample type	qPCR degradation ratio[a]	D19/D18 STR ratio[b]	Average TH01 peak height (RFU)	Identifiler™ peaks detected (%)	nuSRY/nuTH01 STR ratio[b]	DYS393/DYS392 STR ratio[b]	DYS393 peak height (RFU)	Yfiler™ peaks detected (%)
3	Baby Tooth	1.4	1.8	758	100	[c]	N/A	N/A	[c]
5	Muscle	0.74	1.3	762	100	0.54	2.0	1097[d]	100
8	Toothbrush	1.2	2.1	922	100	0.57	4.5	544	100
9	Toothbrush	0.87	2.0	798	100	0.82	2.5	517	100
11	Toothbrush	1.5	2.5	1122	100	0.66	2.3	1830	100
12	Saliva-Coffee Cup	1.3	4.2	897	100	0.86	3.3	978	100
14	Toothbrush	1.0	2.1	846	100	0.94	3.0	916	100
17	Tooth	1.1	1.5	732	100	[c]	N/A	N/A	[c]
18	Toothbrush	1.1	1.5	618	100	0.46	1.9	990	100
19	Bloody BandAid	2.3	1.37	965[e]	100	0.99	2.2	906	100
20	Toenails	5.8	[f]	[f]	[f]	1.2	4.5	862	100
26	Bloodstain Control	1.3	1.2	431[e]	100	[c]	N/A	N/A	[c]
27	Dirty Bloodstain	1.3	1.1	1058	100	[c]	N/A	N/A	[c]
28	Blood on Black Sateen	0.98	1.2	715	100	[c]	N/A	N/A	[c]
29	Blood on Denim	0.77	1.5	729	100	[c]	N/A	N/A	[c]
30	Blood on Redwood Bark-Swabbed	0.83	1.1	575	100	[c]	N/A	N/A	[c]
31	Blood on Redwood Bark-Chipped (1:20)	0.92	1.2	571	100	[c]	N/A	N/A	[c]
32	Blood on Redwood Bark-Chipped (1:200)	1.5	2.5	678	100	[c]	N/A	N/A	[c]
33	Blood in Clodbreaker Soil	3.0	2.8	456[e]	100	[c]	N/A	N/A	[c]
34	Blood in General Ecohumis Soil (1:10)	[g]	[g]	[g]	[g]	[c]	N/A	N/A	[c]
34	Blood in General Ecohumis Soil (1:100)	[g]	[g]	[g]	[g]	[c]	N/A	N/A	[c]
34	Blood in General Ecohumis Soil (1:1000)	[g]	[g]	[g]	[g]	[c]	N/A	N/A	[c]
35	Blood in General Landscape Soil (1:10)	5.0	[g]	[g]	[g]	[c]	N/A	N/A	[c]
35	Blood in General Landscape Soil (1:100)	410[h]	[g]	[g]	[g]	[c]	N/A	N/A	[c]
35	Blood in General Landscape Soil (1:1000)	[g]	[g]	[g]	[g]	[c]	N/A	N/A	[c]
38	Saliva on Wool	1.5	2.7	973[e]	100	[c]	N/A	N/A	[c]
39	Saliva on White Cotton	1.1	1.6	678	100	[c]	N/A	N/A	[c]
41	Saliva on Cigarette Butt	1.9	2.9	630	100	[c]	N/A	N/A	[c]
42	Blood on Newspaper	0.89	1.7	791[e]	100	[c]	N/A	N/A	[c]
43	Blood on Denim	1.2	2.4	884	100	[c]	N/A	N/A	[c]
44	Blood on Wool	1.3	1.3	688	100	[c]	N/A	N/A	[c]
45	Blood on Carpet	1.2	1.2	992[e]	100	[c]	N/A	N/A	[c]
46	Blood on Leather	1.0	1.5	732	100	[c]	N/A	N/A	[c]
49	Semen on Denim	1.3	1.6	1152	100	0.76	1.7	869	100
50	Semen on Wool	1.1	1.6	481	100	0.55	1.5	1171	100
51	Semen on Leather	0.91	1.8	829	100	0.68	2.2	702	100
53	Semen and Blood on Cotton Swab	0.86	[f]	[f]	[f]	0.62	0.92	328	100
55	Semen on Wool	1.6	1.6	822	100	0.68	2.5	1095	100
56	Semen on White Cotton Fabric	1.5	1.8	733	100	0.85	2.9	1023	100
57	Semen on Leather	1.8	1.8	775	100	1.0	2.6	951[d]	100
58	Semen on Denim	2.2	1.9	699	100	0.69	3.0	1235	100
72	Vaginal Swab	2.0	2.6	782	100	[c]	N/A	N/A	[c]
73	Vaginal Swab	1.2	2.1	595	100	0.54	4.4	644	100
74	Vaginal Swab	1.9	2.5	1038	100	[c]	N/A	N/A	[c]
75	Vaginal Swab-EC[i]	1.3	1.6	1488	100	0.0021	3.7	409	100
76	Vaginal Swab-SF[i]	0.96	[j]	ND[e]	26	[c]	N/A	N/A	[c]
77	Vaginal Swab-EC 0 h[i]	0.97	1.6	1201	100	0.0033	4.0	676	100
78	Vaginal Swab-SF 0 h[i]	0.71	[j]	60[e]	13	[c]	N/A	N/A	[c]
79	Vaginal Swab-EC 8 h[i]	0.91	1.6	1248	100	0.00017	[k]	516	94[e]
80	Vaginal Swab-SF 8 h[i]	5.5[h]	[j]	121[e]	62	[c]	N/A	N/A	[c]
81	Male forehead sweat with female saliva	1.0	1.4	915	100	0.020	2.6	1106	100
82	Male underarm sweat with female saliva	1.1	1.4	1091	100	0.00093	ND[e]	ND[e]	0[e]
83	Male saliva on female skin	1.3	1.9	730	100	0.73	1.8	698[d]	100

[a] qPCR degradation ratio = nuCSF quantity/nuTH01 quantity.
[b] STR ratio = peak height of the smallest STR locus/peak height of largest STR locus.
[c] No male DNA detected in this sample.
[d] Data from second injection.
[e] Insufficient DNA remained for optimal STR amplification and the results are from amplification of template quantities estimated to be less than optimal.
[f] Sample consumed in qPCR and Yfiler amplifications.
[g] These soil samples did not amplify via qPCR and STR amplification was not attempted as these samples previously yielded null Identifiler amplifications.
[h] Value skewed due to low nuTH01 quantification value.
[i] Sample is the product of a vasectomized male/female coupling.
[j] The D19 locus was not detected in this sample and, therefore, the D19/D18 STR ratio could not be calculated.
[k] Value could not be calculated as the DYS392 allele was not detected for this sample.

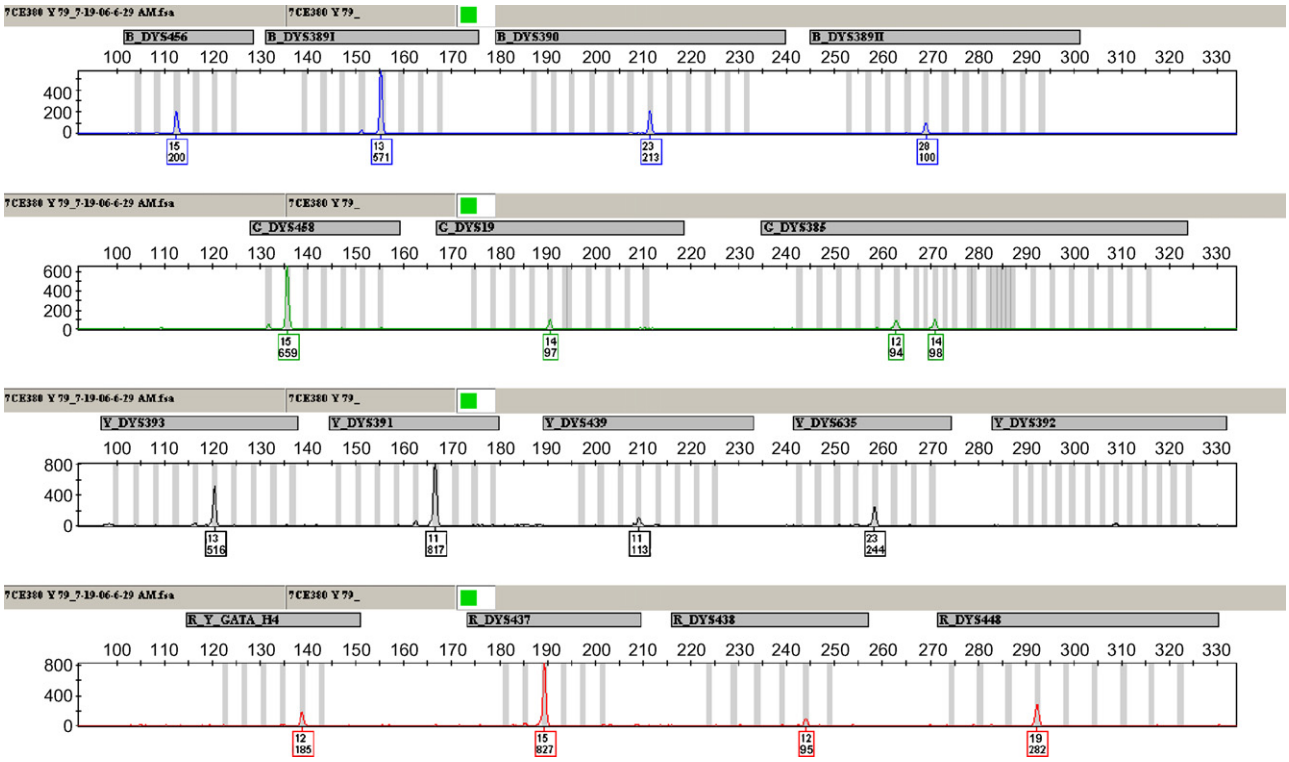


Fig. 5. AmpF*l*STR® Yfiler™ STRs for Item 79 (non-sperm fraction of vaginal swab from vasectomized male/female coupling, 8 h post-coitus). *Note*: STRs are the result of amplifying approximately 300 pg male DNA, less than the normally desired 500 pg male DNA, in the presence of approximately 2000 ng of female DNA.

resulted in partial profiles as indicated in Table 7 for samples 76, 78 and 80. Examination of the STR data from the 1 ng AmpF*l*STR® Identifiler™ amplifications indicates that the quadruplexed nuTH01 target effectively estimates the quantity of DNA in many sample types (Table 7).

3.7. Population studies (SWGDAM 2.7)

The goal of this study was to determine if nuSRY quantifications are consistent with nuTH01 and nuCSF quantifications over a population of male samples. To do this, a total of 225 known male samples (82 African American, 73 Caucasian and 70 Hispanic) were quantified with the nuTH01–nuSRY–nuCSF–IPC qPCR quadruplex assay.

Of the 225 known male samples, 224 yielded signals above threshold at the nuSRY, nuTH01 and nuCSF targets. A single sample yielded a low quantification value, 20 pg, at the nuSRY and nuCSF targets with no DNA detected at the nuTH01 target, which is likely the result of the low quantity of DNA present in this sample. The nuSRY:nuTH01 (male target to long autosomal target ratio), nuSRY:nuCSF (male target to short autosomal target ratio) and nuCSF:nuTH01 (qPCR degradation ratio) ratios were calculated for each ethnic group (Table 8). All 225 known male samples yielded quantification values for the nuSRY target, and none would have resulted in a false negative to indicate a sample devoid of male DNA. While the nuSRY:nuTH01 and nuSRY:nuCSF ratios had average values of approximately 1, minimum values for these ratios were observed to be as low as 0.30 and 0.25, respectively. Deviations of these ratios from the ideal value of 1 are not unexpected given the possibilities of DNA degradation, PCR inhibition, and the intrinsic imprecision associated with each component of the qPCR assay. The exact cause of the low nuSRY:nuTH01 and nuSRY:nuCSF ratios is unknown, but is being investigated by the laboratory.

3.8. Mixture studies (SWGDAM 2.8)

Known male:female DNA mixtures were prepared by mixing male genomic DNA (Promega) with excessive

Table 8
Summary of population study results for single-source male samples

qPCR quantity ratio	African American			Caucasian			Hispanic		
	Mean	Minimum	Maximum	Mean	Minimum	Maximum	Mean	Minimum	Maximum
nuSRY:nuTH01	1.09	0.30	2.13	0.82	0.63	1.00	0.70	0.33	1.00
nuSRY:nuCSF	1.12	0.25	1.76	0.95	0.66	1.32	1.00	0.46	1.46
nuCSF:nuTH01	1.04	0.62	2.71	0.87	0.57	1.24	0.71	0.42	1.06

Table 9
Male:female mixture study results

Mix	Ratio of male to female DNA, 1 to	Male DNA in mix (ng/4 μL)	Female DNA in mix (ng/4 μL)	Number of successful qPCR Amps	Mean nuSRY quantity (ng/4 μL)	Standard deviation nuSRY quantity (ng/4 μL)
1	3,400	0.064	219	3	0.069	0.036
2	6,800	0.032	219	3	0.022	0.015
3	14,000	0.016	219	2	0.019	0.018
4	27,000	0.008	219	3	0.004	0.004
5	6,800	0.064	438	3	0.049	0.027
6	14,000	0.032	438	3	0.039	0.044
7	27,000	0.016	438	2	0.013	0.009
8	55,000	0.008	438	2	0.002	0.001
9	10,000	0.064	657	3	0.041	0.018
10	21,000	0.032	657	3	0.014	0.007
11	41,000	0.016	657	3	0.009	0.011
12	82,000	0.008	657	0	ND	N/A

ND: nuSRY signal not detected.

quantities of HL60 female DNA (ATCC). Each of these mixed DNA samples was then quantitated in triplicate with the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay (Table 9). The nuTH01–nuSRY–nuCSF–IPC qPCR quadruplex assay reproducibly detected as little as 8 pg of male DNA in the presence of 219 ng of female DNA, which represents a 1:27,000 male:female ratio. It should be noted that the variation observed in the nuSRY portion of the quadruplex assay for these samples is not unexpected and is similar to the variation that was observed in the sensitivity study for the samples with low input DNA (44 and 7.2 pg).

3.9. Precision and accuracy (SWGDAM 2.9)

We examined precision by mixing all assay reagents and DNA template in a single tube from which 20 μL portions were uniformly aliquotted into each well of a 96-well plate (4 ng DNA per reaction). The standard deviations in C_T across the plate were 0.066, 0.056, 0.23 and 0.17 cycles for the nuTH01, nuSRY, nuCSF and IPC portions of the assay, respectively. We observed no systematic changes in the mean C_T or standard deviation of C_T across columns or rows (data not shown).

In the absence of an appropriate NIST-traceable human DNA quantification standard, we evaluated ''accuracy'' of quantitation by comparing the results of the quadruplex reproducibility study to results obtained for the same 19 samples quantitated by the nuTH01–nuCSF–IPC *triplex* qPCR assay [18]. The average percent difference between the triplex and quadruplex assays for the nuTH01 measurements for all samples was −4.8% (range of −20% to 11%) and the average percent difference for the nuCSF measurements was −0.66% (range of −23% to 29%) (Table 6).

In addition, we evaluated the quality of the AmpF*l*STR® Identifiler™ STR amplifications and AmpF*l*STR® Yfiler™ amplifications for the same set of 19 samples based on quantifications by the quadruplexed nuTH01 and nuSRY targets, respectively. Using the mean concentration of each sample as determined by the nuTH01 assay from two independent reproducibility runs, 1 ng of each sample, excluding the 9947A sample as indicated below, was used in an AmpF*l*STR® Identifiler™ amplification. For the 9947A, the low concentration of DNA (0.07 ng/μL) resulted in an amplification of 0.7 ng. The average value of the TH01 peaks, using the first injection of each sample, ranged from 713 to 1164 RFU (data not shown). The Promega Human Genomic DNA standards (Male and Female) were amplified and typed to confirm the presence of peaks, but were not further examined as these samples are pooled, rather than single-source, DNA samples. All samples yielded full profiles. Using the mean concentration of each sample as determined by the nuSRY quantitations from two independent reproducibility runs, 0.5 ng from each sample was used in an AmpF*l*STR® Yfiler™ amplification. The average value from the DYS393 peaks, using the first injection of each sample, ranged from 773 to 1638 RFU (data not shown). The Human Genomic Male DNA (Promega) was amplified and typed to confirm the presence of peaks, but was not further examined as this is a pooled DNA sample. All male samples yielded full profiles. These results indicate the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay estimates appropriate template DNA quantities for amplification and detection of autosomal and Y STRs.

3.10. Contamination (SWGDAM 3.6)

The presence of contamination was monitored for with TE^{-4} negative template controls in each experiment with the exception of the plate precision and population studies where the plate configurations did not allow for the addition of a negative template control. A total of 15 TE^{-4} negative template controls were run throughout the validation of the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay, and no nuclear DNA was detected in any of the negative template controls. These results indicate the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay, while sensitive, does not erroneously yield false positives when handled properly even when 45 cycles of two-step qPCR are utilized.

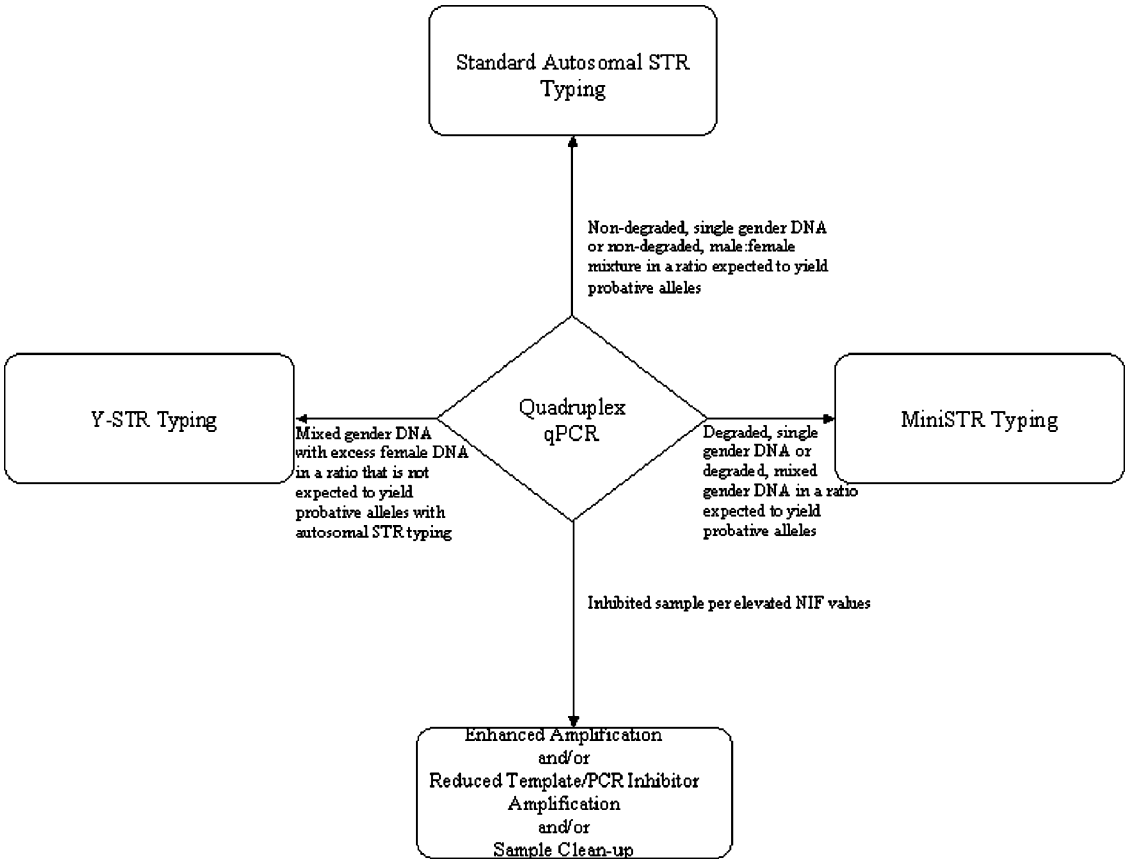


Fig. 6. Flowchart depicting the choice of DNA typing strategy after quantification with the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay.

4. Conclusions

The ability to accurately determine the quantity of DNA present in a forensic sample and simultaneously determine the most appropriate typing system for each sample may result in a substantial increase in laboratory efficiency and a more prudent use of limited samples. The nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay has been shown to provide suitable quantifications for both total human and total human male DNA while providing qualitative information about the level of degradation and the presence of PCR inhibitors. Therefore, analysts equipped with the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay can better judge how to proceed with DNA typing prior to attempting standard autosomal STR typing (Fig. 6). The appropriate use of the ΔIPC and NIF indices in samples containing PCR inhibitors can further increase the first-pass STR typing efficiency and minimize the number of samples that require re-analysis and/or time-consuming interpretation of suboptimal typing results. Finally, the results from the nuTH01–nuSRY–nuCSF–IPC quadruplex qPCR assay can also be used to confirm the results of STR typing, which may prevent unnecessary re-analysis of degraded or inhibited samples that have already yielded the maximum amount of information available.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.fsigen.2007.09.001.

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